soluble antigens and antibodies. All assays for antibody production depend upon the measurement of interaction of elicited antibody with antigen.

Principle

The **principle of ELISA** is very simple. The test is generally conducted in micro titre plates. (Figure 11.2 Micro titre plate).

If the antigen is to be detected the antibody is fixed in the micro titre plate and vice versa. Test sample is added in the microtitre plate, if there is presence of Ag or Ab in the test sample, there will be Ag-Ab reactions (with immobilized Ab or Ag). Later enzyme labelled antibody is added in the reaction mixture, which will combine with either test antigen or Fc portion of test antibody.

The enzyme system consists of:

1. **An enzyme**: Horse Radish Peroxidase(HRP), alkaline phosphatase which is labelled or linked, to a specific antibody.
2. **A specific substrate**:
   - O-Phenyl-diamine-dihydrochloride for peroxidase
   - P Nitrophenyl Phosphate- for Alkaline Phosphatase

Substrate is added after the antigen-antibody reaction. The enzyme hydrolyses the substrate to give a yellow colour compound in case of alkaline phosphatase (Figure 11.3). The intensity of the colour is proportional to the amount of antibody or antigen present in the test sample, which can be quantified using ELISA reader (Figure 11.4 ELISA reader)



Figure 11.2: Micro titre plate



Figure 11.4: ELISA Reader

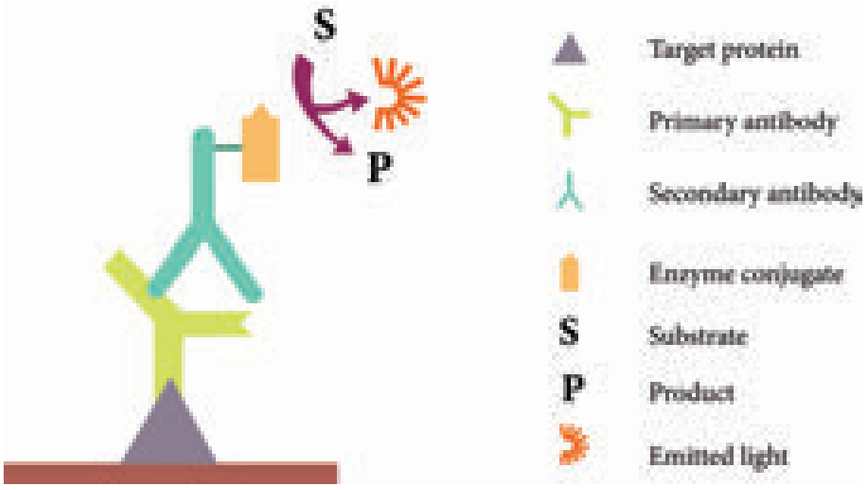


Figure 11.3: Basic steps in ELISA

TN_GOVT_XII_Micro_Biology_CH11.indd 232 04-03-2019 12:31:59